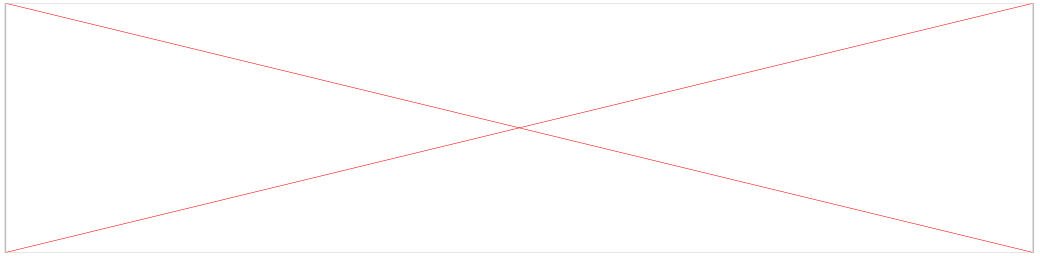
Annual Review for RTC-Tools

TAC annual review, 10 November 2026

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Overview

Open-source optimization framework that helps hydropower operators and energy traders get the most from reservoirs, batteries, and other assets within physical limits and under uncertain forecasts.

Use cases

- Water: model-predictive control of canals, polders and reservoirs; hydropower scheduling; pumped storage
- Energy: battery optimization, power trading, heat networks, multi-energy planning

Links

- Project page
- Documentation
- LFX Insights
- TSC calendar
- Charter
- Repository
- PyPI
- OpenSSF badge
- Roadmap

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Main Repository



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RTC-Tools

Tests & Release passing | TC build success | quality gate passed | coverage 82%

pypi v2.8.1 | downloads/month 5k | downloads 374k | python 3 | License LGPL v3

Health Score Healthy | openssf best practices passing | docs passing

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Incubation criteria: RTC-Tools status

Criterion	Status	Evidence
OpenSSF Best Practices “Passing”	Met	Badge level “passing”
README	Met	README.md
GOVERNANCE	Met	GOVERNANCE.md; Charter; Code of Conduct
TSC roster in LFX	Met	9 members (6 voting, 3 observers); monthly open meetings
COMMITTERS / CODEOWNERS file	Partly	Nine committers listed in CONTRIBUTING; no file yet
Architecture documentation	Gap	Planned for Q1 2027
One-year roadmap	Met	Roadmap (Nov 2026 to Dec 2027), with out-of-scope items
Community growth (LFX Insights)	See slide	Contributions slide
Plan for users and Early Adoption	See slide	Incubation plan slide

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Incubation plan

Step	Closes	Target
COMMITTERS and CODEOWNERS files; align CONTRIBUTING with the LFX TSC roster	Committers file	Dec 2026
Link the posted roadmap from the README	One-year roadmap	Q1 2027
Architecture overview in the documentation	Architecture documentation	Q1 2027
Contributor growth: good-first issues; track organizations in LFX Insights	Community growth	2027
Contributor engagement: bi-weekly coordination meeting with external contributors; monthly open TSC meeting	Community growth	Ongoing
Case study or blog post through the LF Energy marketing form	User community plan	2027
OpenSSF Silver, public adopters list, 18-month release plan (Early Adoption criteria)	Early Adoption	After Incubation

Contributions

RTC-Tools

Nov 5, 2025 → Nov 5, 2026

Organizations leaderboard

All activities

Organization	Total contributions
Stichting Deltares	249 - 68%
Deltares	58 - 16%
PortfolioEnergy	33 - 9%
Katholieke Universiteit Leuven	15 - 4%
LF Release Engineering Actions	8 - 2%
IBA Group	1 - 0%
yacoda	1 - 0%

RTC-Tools

Nov 3, 2025 → Nov 3, 2026

Contributors leaderboard

All activities

Contributor	Total contributions
jarsarasty	706 - 43%
jackvreeken	262 - 16%
Jesus Rodriguez	210 - 13%
k-horvath-deltares	111 - 7%
Jack Vreeken	98 - 6%
Farid Alavi	88 - 5%
Bernhard Becker	67 - 4%
Jorn Baayen	33 - 2%
Klaudia Horvath	15 - 1%
Vedant Madane	13 - 1%

LFX | Insights

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Organizations contributing and/or using in production


















- **Contributing:** LFX lists 12 organizations (Contributors tab), led by Deltares, PortfolioEnergy, KU Leuven, IBA Group and Yacoda; TSC member organizations: Deltares, PortfolioEnergy, Ontario Power Generation
- **Presented their use of RTC-Tools at the 1st User Day** (programme): Ontario Power Generation, US National Weather Service, HydroLogic (Waterschap Rivierenland), PortfolioEnergy, HKV, Romanian Waters, Hydrotec
- **Presenting at the 2nd User Day** (event): PortfolioEnergy, University of Duisburg-Essen, Bundesanstalt für Gewässerkunde, Hydrotec, Manitoba Hydro, Ontario Power Generation, Rijkswaterstaat, RWTH Aachen University

Key Achievements in the past year (1/2)

Area	Achievement	Status
Releases	2.7.2, 2.7.3, 2.8.0, 2.8.1	Nov 2025 to Sep 2026
LP/MILP export	Write linear problems to LP files (#1752)	2.8.0
Custom solver versions	Solver plugin path (#1767); HiGHS solver wheels rtctools-highs, built in rtc-tools-casadi-plugins	2.8.0; wheels 0.1.0 to 0.1.4 on PyPI, May to Jul 2026
Performance and models	Faster delay inlining (#1726) and input mapping (#1732); direct model supply through the pymoca_model hook (#1848)	2.8.0; hook in the next release
Compatibility	uv (#1695), Python 3.10 to 3.14 (#1729), CasADi 3.8 (#1821); Python 3.11 to 3.15 (#1835)	2.8.0; CasADi 3.8 in 2.8.1; Python 3.11 to 3.15 in the next release

Key Achievements in the past year (2/2)

Area	Achievement	Status
LF Energy transition	Move to the rtc-tools organization (#1728); charter, governance and contribution guidelines (2.7.3)	2.7.2, 2.7.3
Security and quality	SonarQube Cloud analysis (#1786); organization-wide SECURITY.md	CI May 2026; policy 2 Nov 2026
Explainability	Tooling that explains optimization results, funded by TKI, with pilot applications at Hoogheemraadschap van Rijnland	In development
QGIS plugin	Build RTC-Tools models in QGIS: network, goals, Modelica and Python generation (rtc_tools_qgis_plugin)	In development; no release yet
Community	1st User Day: 4 Nov 2025, online, 41 participants; 2nd User Day: 9 Nov 2026, online, hosted by LF Energy; bi-weekly contributor meetings and monthly open TSC meetings	4 Nov 2025; 9 Nov 2026; ongoing

Areas the project could use help on

- More visibility (case studies, a blog post or a webinar) to expand the organizational base
- Guidance and support to complete the Incubation requirements and move to the next project phases

Feedback on working with LF Energy

- Positive experience: helpful and collaborative attitude of LF Energy staff and contractors.
- Repository configuration requests have been resolved by LF Energy staff/contractors, but may take longer than necessary. We suggest letting repository maintainers make some basic configuration changes themselves, for example enabling private security reporting.
- Reasons for changes on the LF Energy webpage and project page are not clear or have not been communicated.
- LFX Insights lists Stichting Deltares and Deltares as separate organizations and some contributors under several identities; we would welcome help merging them so that the community metrics are accurate.